

BRANCH DOCUMENT — IRIS-3

Branch: branch_20260512_210035 | Base: main-v1.0

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Decision: GO_WITH_CONSTRAINTS | Confidence: 78%

CHANGE REQUEST

elimina il pannello solare di destra

CASCADE ISSUES IDENTIFIED

[MAJOR] ISSUE-001 — Power generation reduced

Subsystem: solar | Req: L1-PWR-001 | Type: constraint_violation

Removing the right solar panel reduces power generation capacity, potentially exceeding the power budget during imaging mode.

[MINOR] ISSUE-002 — Structural balance altered

Subsystem: mechanical | Req: None | Type: missing_test

Removing the right solar panel alters the structural balance and could affect mechanical stability during launch vibration.

[MINOR] ISSUE-003 — Thermal imbalance risk

Subsystem: thermal | Req: None | Type: reanalysis_required

Removing the right solar panel may introduce thermal imbalances, affecting temperature control and stability.

MISSING ECSS TESTS — NOW REQUIRED

ECSS-E-ST-10-03 §9.2 — Random vibration test with modified solar panel configuration

Reason: To verify the structural integrity and stability after removing the right solar panel.

ECSS-E-ST-10-03 §9.4 — Thermal vacuum test with modified thermal configuration

Reason: To assess the impact on thermal stability and ensure ΔT remains within acceptable limits.

SUBSYSTEM STATUS — UPDATED

Subsystem	Score	Status	Detail	Changed
optical	92%	OK	MTF 0.42 > threshold 0.40	—
thermal	53%	WARN	Thermal imbalance risk	■ MODIFIED
mechanical	80%	OK	Structural balance altered	■ MODIFIED
structure	91%	OK	Mass 33.8 kg / budget 35 kg	—
solar	69%	WARN	Power generation reduced	■ MODIFIED
startracker	67%	WARN	Stray light analysis pending	—
dataproc	71%	WARN	Compression ratio TBC	—

AFFECTED REQUIREMENTS

- L1-PWR-001 — verification evidence invalidated, re-analysis required